

AI Interview Preparation System

Complete Technical Architecture, Implementation Audit & System Specification

Version 1.0.0

Production Verified

Branch: main

Date: October 6, 2026

1. Executive Summary & Project Overview

The **AI Interview Preparation System** (`Prep-Interview`) is a multi-round, full-stack autonomous AI interview simulation platform designed for rigorous screening and technical hiring preparation across **37 technology job roles**.

INTERVIEW ROUNDS

4 Stages

Aptitude, Tech, Coding, HR

JOB ROLES SUPPORTED

37 Roles

Across 6 Tech Domains

QUESTIONS POOL

122+ Verified

Pre-seeded + LLM dynamic

AI ENGINE

Gemini + Rules

Zero-downtime heuristic

Core Interview Workflow

```
User Authentication (Supabase Auth / Local Fallback)
  ↓
Dashboard (Historical tracking, average scores, resume active interviews)
  ↓
Interview Setup Wizard (interview.html)
  |— Step 1: Target Job Role Selection (37 roles across 6 categories with real-time search)
  |— Step 2: Difficulty Level (Beginner / Intermediate / Advanced)
  |— Step 3: Interview Rounds Selection (Aptitude, Technical, Coding, HR)
  ↓
Interview Session Creation (POST /api/interviews → Supabase DB / Memory Store)
  ↓
Round 1: Aptitude & Logic (aptitude.html) — 30 MCQs, timed (30 mins), immediate factual scorecard
  ↓
Round 2: Technical Architecture (technical.html) — 30 questions, voice/text, concept coverage AI eval
  ↓
Round 3: Coding / DSA (coding.html) — 30 C++ algorithmic challenges, approach explanation, 3-state evalu
  ↓
Round 4: HR / Behavioral (hr.html) — 30 questions, STAR framework analysis, voice transcription
  ↓
Finalization (POST /api/interviews/:id/finalize)
  ↓
Cumulative Performance Appraisal & PDF Report (report.html)
```

2. Technology Stack

Layer	Technology	Specification & Usage
Frontend Core	Vanilla ES6+ JavaScript, HTML5	Multi-page application architecture with 11 specialized HTML pages. Zero client-side bundlers required.
Styling & Design	CSS3 Design System	Clean white & deep slate dark theme design tokens, CSS custom properties, responsive Flexbox and Grid.
Backend API	Node.js (v18+) + Express.js	ES Modules (<code>"type": "module"</code>), RESTful endpoints, Helmet security headers, CORS protection, Morgan logging.
Database	Supabase PostgreSQL 15+	Row Level Security (RLS) active on all user data. Fallback in-memory store in <code>memoryStore.js</code> .
AI Engine	Google Gemini API + Heuristic Engine	<code>gemini-1.5-flash</code> / <code>gemini-2.5-flash</code> via <code>@google/genai</code> . Local AST & rule heuristics for zero downtime.
PDF Generation	html2pdf.js + Headless Engine	Client-side vector HTML-to-PDF export and browser print media stylesheets.
Voice / Audio	W3C Web Speech API	SpeechRecognition (voice-to-text dictation) and SpeechSynthesis (Text-to-Speech question reading).

3. Complete Project Directory Structure

```

AI-Interview-Preparation-System/
├─ package.json                # Root NPM package manifest
├─ README.md                   # Project overview
├─ PROJECT_TECHNICAL_DOCUMENTATION.md # Comprehensive technical markdown specification
├─ PROJECT_TECHNICAL_DOCUMENTATION.html # Standalone printable documentation
├─ database/
│   ├─ schema.sql              # PostgreSQL schema, RLS policies, triggers
│   └─ seed_data.sql           # SQL seed scripts for question banks
├─ backend/
│   ├─ server.js               # Express bootstrap & static file server
│   ├─ config/supabase.js      # Supabase client config & fallback validator
│   ├─ controllers/            # Express route controllers (7 domain controllers)
│   ├─ middleware/             # Auth verification & centralized error handler
│   ├─ routes/                 # Express REST API routes
│   ├─ services/               # AI LLM service & Code execution validator
│   ├─ utils/memoryStore.js    # In-memory storage & 122+ pre-seeded questions
│   └─ [16 Automated Test Suites] # Unit & integration test suites
└─ frontend/
    ├─ index.html              # Marketing landing page
    ├─ login.html / signup.html # Authentication pages
    ├─ dashboard.html          # Candidate dashboard & stats
    ├─ interview.html          # 3-step setup wizard
    ├─ aptitude.html           # Round 1: Aptitude & Logic + Scorecard
    ├─ technical.html          # Round 2: Technical Architecture
    ├─ coding.html             # Round 3: Coding / DSA (C++ Only)
    ├─ hr.html                 # Round 4: HR & Behavioral
    ├─ report.html             # Cumulative report & PDF export
    ├─ history.html / profile.html # Past interviews & profile settings
    ├─ preparation.html        # 37-role roadmaps & curricula
    ├─ css/ (12 stylesheets)    # Global tokens & page-specific stylesheets
    └─ js/ (15 modules)         # API client, auth, and round controllers
  
```

4. Detailed Round Implementations

Round 1: Aptitude & Logic (aptitude.html)

- **Format:** 30 Multiple Choice Questions (MCQs) with options A, B, C, D.
- **Timer:** 30-minute global countdown timer with auto-submission upon expiry.
- **Scoring:** Exact deterministic numerical calculation (1 mark per correct answer, 0 for wrong/skipped).

- **Dedicated Result Scorecard:** Instant post-round screen displaying Total Questions, Attempted, Correct, Incorrect, Skipped, Accuracy %, Time Analysis, Topic-wise breakdowns, and step-by-step derivations.

Round 2: Technical Architecture (`technical.html`)

- **Format:** 30 structured conceptual & system design questions.
- **AI Evaluation:** Evaluates candidate answer against expected concepts, categorizing concept coverage, scores out of 10, correct points, missing concepts, technical mistakes, and improvement plans.
- **Voice Support:** Full Web Speech Recognition dictation and SpeechSynthesis reading.

Round 3: Coding / DSA — C++ Only (`coding.html`)

- **Language:** Fixed permanently to **C++ ONLY**.
- **Starter Template:** Strictly clean starter code without exposing reference solutions:

```
#include <iostream>
using namespace std;

int main() {
    // Write your code here

    return 0;
}
```

- **Approach Explanation:** Dedicated candidate approach box to articulate complexity tradeoffs ($O(N)$ time/space).
- **3-State Verdicts:**  **Correct** (88-98 pts),  **Partially Correct** (50-65 pts), and  **Incorrect** (15-30 pts).
- **State Isolation:** Independent state tracking across all 30 problems preserving typed code when switching questions.

Round 4: HR / Behavioral (`hr.html`)

- **Format:** 30 behavioral/leadership scenario questions.
- **STAR Analysis:** AI evaluation structured against Situation, Task, Action, and Result components with KPI metrics.

5. Database Schema & Data Isolation

Table Name	Primary Key	Foreign Keys	RLS Policy & Access
profiles	id (UUID)	auth.users(id)	Users can only view and edit their own profile.
interviews	id (UUID)	user_id → auth.users(id)	Full CRUD restricted to auth.uid() = user_id .
interview_answers	id (UUID)	interview_id → interviews(id) user_id → auth.users(id)	Insert & select restricted to auth.uid() = user_id .
interview_results	id (UUID)	interview_id → interviews(id) user_id → auth.users(id)	Unique per interview; restricted to owner.
aptitude_questions	id (UUID)	None	Read-only for authenticated users; service role write.
technical_questions	id (UUID)	None	Read-only for authenticated users; service role write.
coding_questions	id (UUID)	None	Read-only for authenticated users; service role write.
hr_questions	id (UUID)	None	Read-only for authenticated users; service role write.

6. API Endpoints Reference

Method	Endpoint	Description	Auth Required
GET	/api/health	System diagnostics & provider status	No
POST	/api/interviews	Create new interview session	Yes
GET	/api/interviews	List authenticated user interviews	Yes
GET	/api/interviews/:id	Get interview session, answers, results	Yes
PATCH	/api/interviews/:id/progress	Update round index / status	Yes
POST	/api/interviews/:id/finalize	Finalize appraisal & calculate scores	Yes
GET	/api/aptitude/questions	Retrieve aptitude question pool	Yes
POST	/api/aptitude/answers	Submit & record aptitude answer	Yes
GET	/api/technical/questions	Retrieve technical question pool	Yes
POST	/api/technical/evaluate	Evaluate technical conceptual answer	Yes
GET	/api/coding/questions	Retrieve coding challenge pool	Yes
POST	/api/coding/submit	Submit C++ code for 3-state evaluation	Yes
GET	/api/hr/questions	Retrieve behavioral questions	Yes
POST	/api/hr/evaluate	Evaluate STAR behavioral answer	Yes
GET	/api/profile	Get profile info & career statistics	Yes
PUT	/api/profile	Update user career profile	Yes

7. Summary & Safety Matrix

This documentation represents the exact, fully verified state of the AI Interview Preparation System codebase. Future modifications should consult the Future Change Safety Map in

`PROJECT_TECHNICAL_DOCUMENTATION.md` before altering coding state isolation, round aggregation, or authentication middleware.
